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1) Of course, first we need to mount the disk: `mount /dev/sr0 /mnt/`. Then we can create repositories:

First, we go to `/etc/yum.repos.d` and create two files there named `BaseOS.repo` and `AppStream.repo`.

For `BaseOS.repo`:

```
[root@rhel10 /etc/yum.repos.d #] cat BaseOS.repo
[BaseOS.repo]
name=BaseOS repository
baseurl=file:///mnt/BaseOS
enabled=1
gpgcheck=0
```

For `AppStream.repo`:

```
[root@rhel10 /etc/yum.repos.d #] cat AppStream.repo
[AppStream.repo]
name=AppStream Repo
baseurl=file:///mnt/AppStream
enabled=1
gpgcheck=0
```

I swear, I did not copy it from anywhere ☺

2) I can show some steps to check deeply whether the repo is working properly or not:

- `ll /mnt/` - To check whether we mount the disk successfully or not.
- `ll /etc/yum.repos.d` - To check whether we create repositories successfully or not.
- `cat /etc/yum.repos.d/<REPO_NAME>.repo` - To see whether we add all mandatory fields successfully or not.
- `yum repolist` - It is a yum command to see all repositories. However, it is not always reliable, so the previous stages might be better.
- `yum update` - It is also a good step to check. If update process is not successful, there is a problem.

3) We can run this command:

```
[root@rhel10 ~ #] yum list installed | grep "httpd"
```

From the internet, I have found a better way to determine it:

```
[root@rhel10 ~ #] rpm -q httpd  
package httpd is not installed
```

4) Currently `httpd` is not installed in my system. So firstly, I will install it:

```
[root@rhel10 ~ #] yum install httpd -y  
Last metadata expiration check: 0:17:20 ago on Mon 09 Mar 2026 05:48:11 AM +04.  
Dependencies resolved.  
=====
```

Package	Architecture
=====	
Installing:	
httpd	x86_64
Installing dependencies:	
apr	x86_64
apr-util	x86_64
apr-util-ldb	x86_64
httpd-core	x86_64
httpd-filesystem	noarch
httpd-tools	x86_64
redhat-logos-httpd	noarch
Installing weak dependencies:	
apr-util-openssl	x86_64
mod_http2	x86_64
mod_lua	x86_64

```
=====
```

After installing it, I remove it:

```
[root@rhel10 ~ #] yum remove httpd -y
Dependencies resolved.
=====
Package                                         Architecture
=====
Removing:
httpd                                           x86_64
Removing unused dependencies:
apr                                             x86_64
apr-util                                       x86_64
apr-util-ldap                                  x86_64
apr-util-openssl                              x86_64
httpd-core                                     x86_64
httpd-filesystem                             noarch
httpd-tools                                   x86_64
mod_http2                                     x86_64
mod_lua                                        x86_64
redhat-logos-httpd                           noarch

Transaction Summary
```

So, our commands for these processes are:

```
`yum install httpd -y`
```

```
`yum remove httpd -y`
```

(It is in my local repository, no worry)

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- 5) We can use either ``yum search <PACKAGE_NAME>`` or ``yum whatprovides <PACKAGE_NAME>`` in that kind of case. However, ``whatprovides`` option is better, because it searches much deeper than the ``search`` one.

```
[root@rhel10 ~ #] yum whatprovides netstat
AppStream Repo
BaseOS repository
net-tools-2.0-0.73.20160912git.el10.x86_64 : Basic networking tools
Repo      : @System
Matched from:
Filename   : /usr/bin/netstat

net-tools-2.0-0.73.20160912git.el10.x86_64 : Basic networking tools
Repo      : BaseOS.repo
Matched from:
Filename   : /usr/bin/netstat
```

6) After some research I found that we need to use `--hidden` option to get that.

With the option, we get all groups including hidden ones:

```
[root@rhel10 ~ #] yum grouplist | wc #Normal version
   24    67   588
[root@rhel10 ~ #] yum grouplist --hidden | wc #Full version with --hidden
   79   201  1831
```

7) First, I searched for this group to find its full name:

```
[root@rhel10 ~ #] yum grouplist --hidden | grep -i "web server"
Basic Web Server
```

Now it is time to install this group:

```
[root@rhel10 ~ #] yum groupinstall "Basic Web Server" -y
Last metadata expiration check: 0:22:25 ago on Tue 10 Mar 2026 12:14:18 PM +04.
Dependencies resolved.
```

Package	Architecture	Version
Installing group/module packages:		
httpd	x86_64	2.4.63-4.el10
httpd-manual	noarch	2.4.63-4.el10
mod_fcgid	x86_64	2.3.9-35.el10
mod_ssl	x86_64	1:2.4.63-4.el10
Installing dependencies:		
apr	x86_64	1.7.5-2.el10
apr-util	x86_64	1.6.3-21.el10
apr-util-ldb	x86_64	1.6.3-21.el10
httpd-core	x86_64	2.4.63-4.el10
httpd-filesystem	noarch	2.4.63-4.el10
httpd-tools	x86_64	2.4.63-4.el10
redhat-logos-httpd	noarch	100.2-1.el10
Installing weak dependencies:		
apr-util-openssl	x86_64	1.6.3-21.el10
mod_http2	x86_64	2.0.29-3.el10
mod_lua	x86_64	2.4.63-4.el10
Installing Groups:		
Basic Web Server		

(No need to show all installation process)

-
- 8) We can visit docker official website to reach information about its installation process. So, in the first step we need to remove some packages from the system to be able to download docker successfully.

```
[root@rhel10 ~ #] sudo dnf remove docker \
docker-client \
docker-client-latest \
docker-common \
docker-latest \
docker-latest-logrotate \
docker-logrotate \
docker-engine \
podman \
runc
```

`yum` might report that you have none of these packages installed.

Now it is time to add docker repo to `/etc/yum.repos.d/`:

```
[root@rhel10 ~ #] yum config-manager --add-repo https://download.docker.com/linux/rhel/docker-ce.repo
Adding repo from: https://download.docker.com/linux/rhel/docker-ce.repo
[root@rhel10 ~ #] cd /etc/yum.repos.d
[root@rhel10 /etc/yum.repos.d #] ll
total 24
-rw-r--r--. 1 root root 88 Mar 9 05:47 AppStream.repo
-rw-r--r--. 1 root root 78 Mar 8 02:21 BaseOS.another
-rw-r--r--. 1 root root 85 Mar 9 05:47 BaseOS.repo
-rw-r--r--. 1 root root 795 Mar 11 23:43 docker-ce.repo ←
-rw-r--r--. 1 root root 129 Mar 8 02:29 mnt_AppStream.another
-rw-r--r--. 1 root root 376 Mar 7 23:25 redhat.repo
```

We can use `cat` command to its contents:

```
[root@rhel10 /etc/yum.repos.d #] cat docker-ce.repo
[docker-ce-stable]
name=Docker CE Stable - $basearch
baseurl=https://download.docker.com/linux/rhel/$releasever/$basearch/stable
enabled=1
gpgcheck=1
gpgkey=https://download.docker.com/linux/rhel/gpg

[docker-ce-stable-source]
name=Docker CE Stable - Sources
baseurl=https://download.docker.com/linux/rhel/$releasever/source/stable
enabled=0
gpgcheck=1
gpgkey=https://download.docker.com/linux/rhel/gpg

[docker-ce-test]
name=Docker CE Test - $basearch
baseurl=https://download.docker.com/linux/rhel/$releasever/$basearch/test
enabled=0
gpgcheck=1
gpgkey=https://download.docker.com/linux/rhel/gpg

[docker-ce-test-source]
name=Docker CE Test - Sources
baseurl=https://download.docker.com/linux/rhel/$releasever/source/test
enabled=0
gpgcheck=1
gpgkey=https://download.docker.com/linux/rhel/gpg
```

As you can see, it has added four sources for docker, but just one of them is enabled by default.

Now we can install the packages:

```
[root@rhel10 /etc/yum.repos.d #] yum install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Docker CE Stable - x86_64
Dependencies resolved.
```

Package	Architecture	Version	Repository
Installing:			
containerd.io	x86_64	2.2.1-1.el10	docker-ce-stabl
docker-buildx-plugin	x86_64	0.31.1-1.el10	docker-ce-stabl
docker-ce	x86_64	3:29.3.0-1.el10	docker-ce-stabl
docker-ce-cli	x86_64	1:29.3.0-1.el10	docker-ce-stabl
docker-compose-plugin	x86_64	5.1.0-1.el10	docker-ce-stabl
Installing dependencies:			
fuse-overlayfs	x86_64	1.15-1.el10	AppStream.repo
Installing weak dependencies:			
docker-ce-rootless-extras	x86_64	29.3.0-1.el10	docker-ce-stabl

```
Transaction Summary
Install 7 Packages

Total size: 100 M
Total download size: 100 M
Installed size: 394 M
Is this ok [y/N]: y
```

(I do not show full installation process, because it is very long)

Now for fun, we can enable docker service and pull one image from there:

```
[root@rhel10 /etc/yum.repos.d #] systemctl enable --now docker
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.
[root@rhel10 /etc/yum.repos.d #] docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[root@rhel10 /etc/yum.repos.d #]
```

9) If you wanted me to remove the docker packages, I just need to run this command:

```
[root@rhel10 ~ #] yum remove docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin docker-ce-rootless-extras
Dependencies resolved.
```

Package	Architecture	Version	Repository
Removing:			
containerd.io	x86_64	2.2.1-1.el10	@docker-ce-stable
docker-buildx-plugin	x86_64	0.31.1-1.el10	@docker-ce-stable
docker-ce	x86_64	3:29.3.0-1.el10	@docker-ce-stable
docker-ce-cli	x86_64	1:29.3.0-1.el10	@docker-ce-stable
docker-ce-rootless-extras	x86_64	29.3.0-1.el10	@docker-ce-stable
docker-compose-plugin	x86_64	5.1.0-1.el10	@docker-ce-stable
Removing unused dependencies:			
fuse-overlayfs	x86_64	1.15-1.el10	@AppStream.repo
Transaction Summary			

Then just run:

```
[root@rhel10 ~ #] sudo rm -rf /var/lib/docker
sudo rm -rf /var/lib/containerd
```

If you want to remove docker repository, too:

```
[root@rhel10 /etc/yum.repos.d #] rm -rf docker-ce.repo
```

There is another way to remove installed packages. So, we can use `dnf history` command and it shows last dnf usages. Each usage has its own id. By using the id, we can run `dnf history undo <ID>`. Now the installed packages in process with this id will be removed now:

```
[root@rhel10 /etc/yum.repos.d #] dnf history
```

ID	Command line
13	install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin


```
[root@rhel10 /etc/yum.repos.d #] dnf history undo 13
Last metadata expiration check: 0:02:27 ago on Thu 12 Mar 2026 12:07:37 AM +04.
Dependencies resolved.
```

Package	Architecture	Version
Removing:		
containerd.io	x86_64	2.2.1
docker-buildx-plugin	x86_64	0.31
docker-ce	x86_64	3:29
docker-ce-cli	x86_64	1:29
docker-ce-rootless-extras	x86_64	29.3
docker-compose-plugin	x86_64	5.1.0
Removing dependent packages:		
fuse-overlayfs	x86_64	1.15
Transaction Summary		

10) We should explain to the system that whenever we run proper command to install any package (like `yum install nano`), it should search it from this specific directory (mountpoint).

When we run yum command, it does not search for the full system, it comes to repositories and find the proper locations from here. The directory to put the repositories in is `/etc/yum.repos.d`.

We create a file here, a repository with the extension of `.repo` for package directories like `AppStream` or `BaseOS`.

11) A **dependency** is a package/software that another package **requires to function**.

Dependency hell is when managing dependencies becomes very **difficult and exhausting** due to conflicts, chains, or version mismatches.